

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

9 Further Human–Computer Interaction (afb21)

British houses are often wastefully heated even when there is nobody home. The problem is that it takes a long time to heat up to a comfortable temperature, and it is too difficult to program thermostats to account for the variable routines an occupant might have from one day to the next.

Imagine someone who owns an Internet-connected thermostat, a semi-autonomous car, and a voice-activated smart speaker. A generative language model fine-tuned for dialog, like the recently popular ChatGPT, could be used via the smart speaker to define a policy such that, on cold days, the heating will come on when the car will be returning home in about 30 minutes.

- interaction space (a) Complete the following voice dialog that might be used by this person to define the above heating policy for the next year.

USER: I want to define a new policy to control the
central heating

BOT: What is the new policy?

USER: If the outside temperature is less than 10 degrees,
use a special winter policy

BOT: What is the special winter policy?

[2 marks]

Answer: USER: The heating should come on the when the car is near home

BOT: What is near home?

USER: Ask the car for estimated time that I will arrive. If the estimated time is less that 30 minutes in the future, turn on the heating.

BOT: I have changed the policy.

USER: In one year from today, go back to the old policy.

BOT: OK.

- waves of HCI (b) Which wave of HCI theory is most relevant to this interaction scenario, and why? [2 marks]

Answer: The scenario relates mainly to mechanical automation, carried out by a single user, so is most like first-wave HCI. It was more typical for first-wave user interfaces to be based on control panels and plain text rather than speech. Chatbots are more of an entertainment technology, which is characteristic of third wave, so there is a mismatch between the interaction style and the task..

- cognitive dimensions (c) Suggest *four* cognitive dimensions that are relevant to the dialog. [4 marks]

Answer: visibility, hidden dependencies, premature commitment, progressive evaluation

cognitive
dimensions

(d) For *each* dimension, explain:

- (i) How this dimension is relevant to the voice dialog.
- (ii) What its significance is in relation to the user's goals.
- (iii) How a voice dialog system might behave differently to improve usability on this dimension.

[12 marks]

Answer: (i) Visibility is an obvious challenge for voice interaction, because the state changes resulting from interaction are not directly visible. The “notation” is a transient medium that cannot be inspected visually, but only queried. (ii) When defining policy for future system behaviour, as in this case, the user must maintain their own mental model of specified behaviour, which they can confirm by querying. (iii) The most obvious improvement would be to add a visual interface with a notational representation. If interaction must be audio-only, then the policy could be output in speech form, perhaps with navigation by query to refer to specific parts of it.

Hidden dependencies: (i) With no visibility, all dependencies will be hidden, so there is a trivial relationship between these two dimensions. Assuming that some kind of voice-based query is used to report the current policy, this might include queries such as “confirm the definition of near home”. (ii) The response to that query would not necessarily include the implications of other things that the definition affects (for example, we can imagine that very cold weather would also impact driving times, resulting in unexpected relationships between the car navigation and the home heating system). (iii) One way to ameliorate this might be to include a search function, or a simple proof engine, that would trace dependencies and then report them, in response to “debugging” command such as “tell me the implications of a change to the definition of near home”. A further problem of hidden dependencies, in systems based on large language models such as GPT, is that there may be other statistical relations between the words used, that are encoded in the language model, but might not be apparent to the user.

Premature commitment: (i) This has traditionally been a problem with voice interaction, because the order of utterances in time is preserved in the dialog. In contrast, text-based program editors allow random access to any part of the text, so that it can be constructed in a different sequence. (ii) Premature commitment is not too much of a problem if the order of utterances is “natural” (according to the mental model of the user), but the above dialog already demonstrates some ways in which the user has to anticipate later steps, such as defining the need for a “special winter policy”, to avoid over-writing other aspects of the heating control. Premature commitment is exacerbated by high viscosity, and in this dialog the user has anticipated this by use of special labels so that they can, for example, cancel the special winter policy, or even modify it. (iii) The dialog I have provided already shows some example of how usability can be improved by allowing the system to take some initiative - here by asking clarification questions such as “what is near home”, allowing references to be made to undefined terms. Some degree of laziness could be added here, so that the user can delay defining a term until they have identified more of the context variables that could be relevant to the policy.

Progressive evaluation: (i) Progressive evaluation involves some kind of execution, which might be straightforward if the policy only changes the behaviour of the dialog system itself, but more challenging if there are side effects in the real world. (ii) In the example scenario, it is quite difficult for the user to evaluate parts of the policy, if this would involve waiting for the outside temperature to fall within the specified range, and for the car to be in a particular

location. (iii) In a voice dialog, the most likely way to provide progressive evaluation would be either to link the dialog to a world simulation (a digital twin) that could be inspected by the user to assess outcomes of the policy, or else to enhance the dialog with support for counterfactuals, so that the user is able to test the policy with “what if” or “assume that” questions. Unfortunately this is quite challenging for large language models at present, as shown by experiments in which prompts must be used to explicitly manipulate the dialog state, for example in a request to forget all prior interaction.
